Week 08 Assignment: Level Streaming and Game System Organization

Brief

In professional game development, levels are rarely isolated spaces. Instead, games are structured as interconnected areas where systems and rules must be organized clearly so they can scale as the project grows.

This week, your goal is to expand your game by adding an additional level and connecting it to your existing environment. You will also begin reorganizing your Blueprint logic so that gameplay rules and persistent data are handled by the appropriate Unreal systems.

As projects become more complex, it becomes important to move logic out of the **Level Blueprint or Player Blueprint** and into systems such as **Game Mode** (for level-specific rules) and **Game Instance** (for data that persists across levels).

Your focus this week is on **expanding your playable space and improving the organization of your gameplay systems**.

Requirements

1. Create a New Level or Streamed Area

Add **at least one new playable space** to your project.

This can be implemented in one of the following ways:

- A new **Level** connected using a portal, exit, or trigger
- A **Level Streaming** setup where a new sublevel loads when the player enters an area

Your new space should have a **clear purpose or identity**. Examples include:

- A challenge room
- A secret area
- A puzzle space
- A new biome or environment
- A reward room or goal area

The player must be able to **transition between the original level and the new area**.

2. Implement a Level Transition

Create a working method for the player to move between spaces.

Examples include:

- A portal or doorway that opens another level
- A trigger box that loads a streamed level
- An exit that returns the player to the original area

The transition should work consistently and allow the player to continue playing the game.

3. Refactor Gameplay Logic

Move **at least one gameplay system** out of the **Level Blueprint or Player Blueprint** and into a more appropriate system.

Examples include:

Game Mode (Level Rules)

Good candidates include:

- Score rules
- Level objectives
- Spawn behavior
- Win/lose conditions
- Level-specific mechanics

Game Instance (Persistent Data)

Good candidates include:

- Total score across levels
- Inventory or collected items
- Progress tracking
- Score multipliers
- Persistent player stats

The goal is to demonstrate that you are beginning to **separate gameplay systems logically** rather than placing all logic in a single Blueprint.

Submission Deliverable

Upload a **video screen recording (2–4 minutes)**showing the following:

1. Gameplay Demonstration

Show the player moving from the original level into the new level or streamed area. Demonstrate that the transition works and that the new space can be explored.

2. Blueprint System Organization

Open the Blueprint(s) where you reorganized your logic and briefly show:

- The **Game Mode** or **Game Instance** Blueprint
- The variables or functions that were moved there
- Where those systems are being accessed from other Blueprints

3. Explanation of Changes

Briefly explain:

- What new level or area you added
- How the player transitions between spaces
- What gameplay logic you moved and why

Voice explanation is optional but encouraged.

Evaluation Criteria

Your submission will be evaluated based on the following:

Level Expansion

A new playable area has been added and can be accessed from the original level.

Functional Transition

The player can reliably move between levels or streamed spaces.

System Organization

Gameplay logic has been moved into an appropriate system such as Game Mode or Game Instance.

Clarity of Explanation
Your video clearly demonstrates the new area and explains the changes you implemented.

Goal

This assignment is designed to help you move from a **single-level prototype** toward a **more scalable game structure**, where environments, rules, and systems are organized in a way that supports larger projects.